

Raw Log Triage Pipeline

Any unstructured log → validated, structured JSON — on a local model.

THE CORE IDEA · WHAT MAKES IT UNIQUE

Unstructured logs in. **Structured, validated JSON out.**

Format-adaptive: one tool reads Linux syslog, HDFS, Android — or any format it has never seen — with zero per-format parsing code.

UNSTRUCTURED — raw, mixed-format log

```
Jun 14 sshd: check pass; user unknown
Jun 9 kernel: PCI: Found IRQ 5
Jun 15 sshd: authentication failure;
Jun 30 combo sshd: Authentication failed
      from 163.27.187.39: Permission denied
Jun 9 kernel: PCI: Found IRQ 11
```



STRUCTURED — validated JSON

```
{
  "service_name": "sshd",
  "timestamp": "Jun 30 ...",
  "error_severity": "fatal",
  "remediation": "Review...",
  "occurrence_count": 1
}
```

How it works — two-stage pipeline

```
$ python pipeline.py app.log --provider ollama|google --model gemma-4-26b-a4b-it -o out.json
```

01 · DEDUP

Ingest & templize

`dedup_templates()` masks PIDs, IPs, hex, numbers & key=value, collapsing thousands of near-identical lines into unique templates with an `occurrence_count`.



02 · PROFILE

Stage 1 — Profile

1 model call on the top templates returns a JSON profile: format, service & timestamp rules, severity scheme, and benign patterns (auto-dropped).



03 · TRIAGE

Stage 2 — Triage

50 templates / call with system prompt = profile + a hard-coded output contract (5 keys, severity rules). The contract is never model-generated.



04 · VALIDATE

Validate & dedupe

`validate_entries()` enforces the 5 keys, coerces bad severities to "error", dedupes by (service, timestamp, source_line) → final JSON.



Templates, not raw lines

500 → 26

raw HDFS lines → templates · ~14 calls, not 250+

The pipeline reasons over deduplicated templates, not raw lines. Masking the variable parts — PIDs, IPs, hex, numbers, key=value — collapses thousands of near-identical lines into a handful of unique patterns, so the model sees each pattern once. An **occurrence_count** preserves how often each fired.

Engineering details



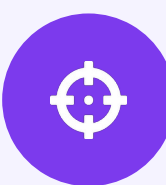
"think": false

Disables chain-of-thought so local Gemma returns JSON immediately instead of hanging on reasoning.



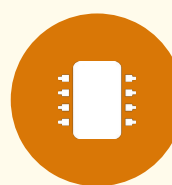
Retry on bad JSON

One stricter-prompt retry before a malformed chunk is skipped — resilient to flaky model output.



source_line in every entry

Cites the exact offending line — fulfilling the hackathon's "locate the precise line" requirement.



Gemma-4 · Google AI Studio

gemma-4-26B-A4B-it for final evaluation; Ollama for local runs. Gemma-only, per rules.